

1. We recall that the limit $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ using the polar coordinates r, θ with $x = r \cos \theta$ and $y = r \sin \theta$, is obtained as follows.
- (i) We first find the limit along some known lines passing through the point $(0,0)$ in order to identify the candidate limit L .
 - (ii) We estimate $|f(r \cos \theta, r \sin \theta) - L| \leq g(r)$, where g is a function of r only with the limit $\lim_{r \rightarrow 0^+} g(r) = 0$. The function g is obtained through our knowledge of the properties of the trigonometric functions sine and cosine.
 - (iii) We then say that $\lim_{r \rightarrow 0^+} f(r \cos \theta, r \sin \theta) = L$ uniformly in θ which is equivalent to saying that $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = L$.

Find the limit $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ for:

$$(a) \quad f(x,y) := \frac{\ln(1+xy)}{\sqrt{x^2+y^2}} \qquad (b) \quad f(x,y) := \frac{x^2y^3}{x^4+y^4}.$$

Hint: You can use $\cos^4(t) + \sin^4(t) \geq 1/2$ in (b).

$$2. \text{ Consider the function } f(x,y) := \begin{cases} \frac{\arcsin(x^2+y^2)}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

- (a) Find the domain of the function f .
 - (b) Study the continuity of f at the point $(0,0)$.
 - (c) Find, if they exist, $f_x(0,0)$ and $f_y(0,0)$.
 - (d) Study the differentiability of the function f at $(0,0)$.
3. Let $f(x,y) = x^2y^3$.
- (i) Show that vector $\vec{n} = (2, 3, -1)$ is perpendicular to the tangent plane to the graph of f at the point $(1, 1, 1)$.
 - (ii) Find the points (x, y, z) on the graph of f at which the tangent plane is perpendicular to the vector $\vec{u} = (-1, 3, 1)$
(Hint: What is the expression of the normal vector to the tangent plane at a point (x, y, z) on the graph of f ?).

4. Let $\lambda \in \mathbb{R}$. Consider the function $f(x, y) := \begin{cases} \frac{x^2 \sin y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ \lambda & \text{if } (x, y) = (0, 0). \end{cases}$

- (a) Find the value of the parameter λ such that the function f is continuous at $(0, 0)$.
- (b) Is the function f differentiable at $(0, 0)$ for the value of λ obtained in (a)?

5. Given the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = y \arctan(x + y)$ for every $(x, y) \in \mathbb{R}^2$.

- (a) Write down the cartesian equation of the tangent plane to the graph of f at the point $(0, -1)$.
- (b) Write down the linear approximation of the function f around the point $(0, -1)$.
- (c) Deduce from (b) an approximation of the number $f(-0.09, -0.9)$.